

Project Initialization and Planning Phase

Date	1 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

The **OptiCrop – Smart Agricultural Production Engine** project focuses on understanding the challenges faced by farmers while selecting suitable crops. This customer problem statement helps identify farmers' needs, understand their difficulties, and design an AI-based crop recommendation system that improves agricultural productivity and decision-making.

A well-defined problem statement enables the project to recommend the most suitable crop based on soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH, and rainfall. By understanding the farmer's perspective, the system provides accurate crop recommendations, reduces the risk of crop failure, and supports sustainable farming practices.

Example:



Problem Statement (PS)	I am (Customer)	I’m trying to	But	Because	Which makes me feel
PS-1	A farmer	Select the most suitable crop for my farmland	I don't know which crop is best for my soil and weather conditions	Soil nutrients and climate vary from season to season	Confused and worried about crop failure